

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Technical Presentation

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 3 – Technical Presentation
Weightage:	35% of CIA	Total Marks:	100
CO1 Statement:	Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.		
Student Name:	G. Santosh Kumar	Reg. No.:	192524020
Section / Batch:	2025 Batch	Date:	15/07/2026

Code of Conduct

I G. Santosh Kumar _____ (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G. Santosh Kumar _____

Instructions

- This assessment contains technical presentation. Total: 100 Marks.
- When a student delivers a technical presentation on a data structures topic, evaluation should be based on the following requirements: Concept explanation, Real-world application and Clarity of presentation.
- Demonstrate clear understanding, not just reading slides
- Use of tools: PowerPoint / Google Slides.
- Duration: 7–10 minutes presentation + 3 minutes Q&A.

Section / Topic	Focus	Q Nos	Marks
Data Structure	Real-Time Applications	1	100
GRAND TOTAL:		100 Marks	

Annexure A – Technical Presentation

A web browser supports navigating backward through previously visited pages. Recommend a suitable ADT and prepare a technical presentation explaining how it improves performance efficiency. (100 Marks)

(OR)

A music streaming application maintains a playlist where songs can be added, removed, or reordered dynamically. Recommend a suitable data structure and justify based on flexibility and efficiency. Prepare a technical Presentation. (100 Marks)

(OR)

A metro rail network maps stations and routes to determine reachable stations from a source station. Suggest a suitable data structure for representation and justify your answer. (100 Mark)

(OR)

A calculator application evaluates arithmetic expressions entered by users. Recommend a suitable data structure and justify based on expression processing efficiency. (100 Marks)

(OR)

A cafeteria token system serves customers strictly in the order of arrival. Identify the suitable ADT and justify why it ensures fairness. (100 Marks)

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	20	Demonstrates deep and accurate understanding of data structure with insights and connections	Shows clear understanding with minor gaps	Basic understanding; some misconceptions present	Limited or incorrect understanding
Content Organization	30	Logical, well-structured, smooth flow; strong introduction and conclusion	Mostly organized with clear flow	Some organization but lacks clarity or transitions	Disorganized, difficult to follow
Technical Depth	20	Includes algorithms, complexity analysis, and advanced insights	Includes algorithm and basic complexity analysis	Limited technical details; missing depth	Minimal or no technical content
PowerPoint Slides	20	Excellent design, clear visuals, enhances understanding	Good slides with minor issues	Basic slides; somewhat cluttered or unclear	Poor slides or no visual support
Communication Skills	10	Clear, confident, engaging delivery; excellent articulation	Clear delivery with minor hesitation	Some hesitation; unclear in parts	Poor communication; difficult to understand

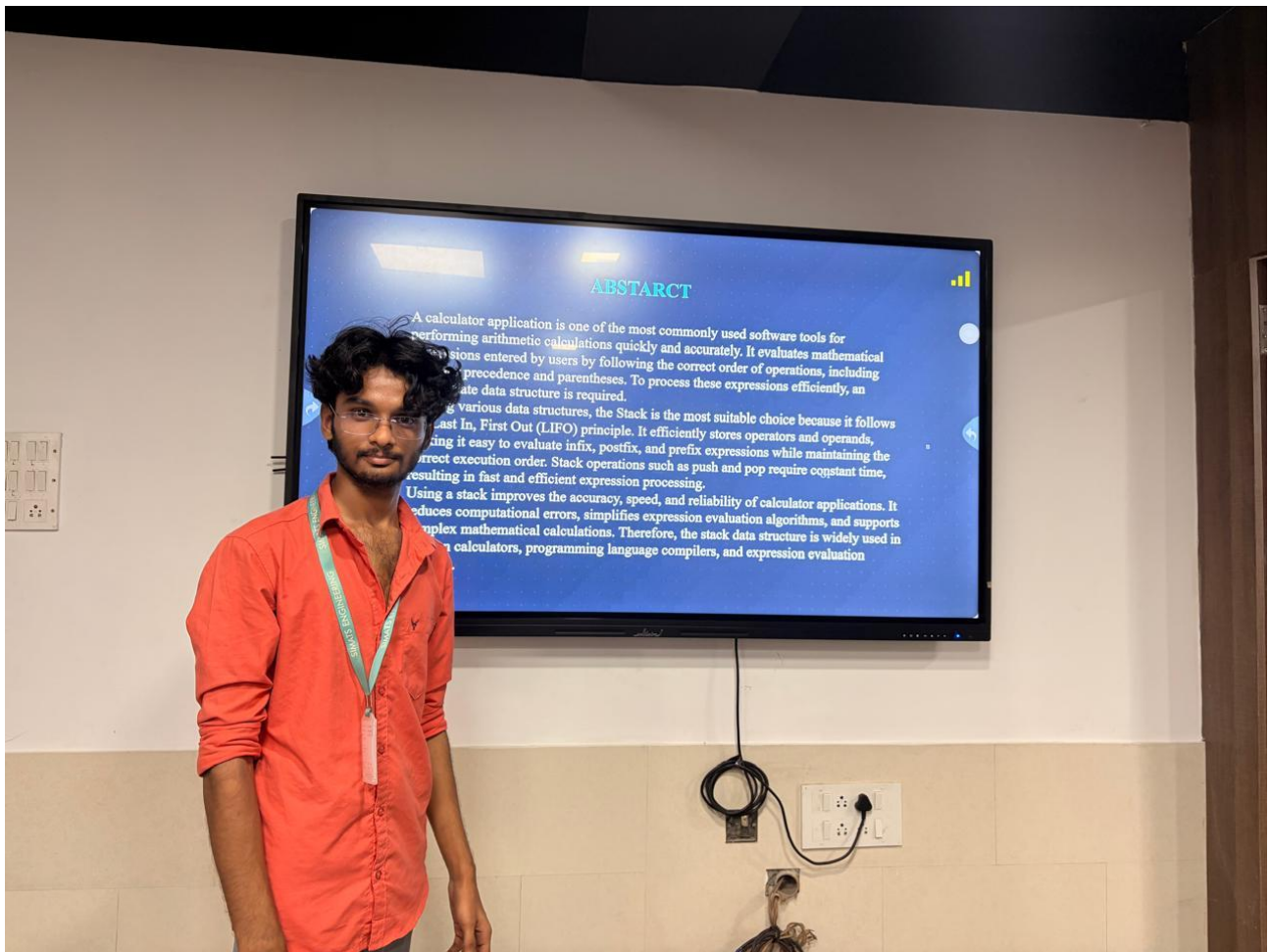
Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

INTRODUCTION

- A calculator application is a software program that performs arithmetic operations such as addition, subtraction, multiplication, and division based on user input.
- It evaluates mathematical expressions by following the correct order of operations, including operator precedence and the use of parentheses
- Efficient expression evaluation requires an appropriate data structure to store and process operators and operands during computation.
- The Stack data structure is commonly used because it follows the Last In, First Out (LIFO) principle, making expression processing simple and efficient.

OBJECTIVES

- Understand how arithmetic expressions are processed.
- Select a suitable data structure for expression evaluation.
- Improve execution speed and efficiency.
- Handle operator precedence correctly.
- Support parentheses and complex mathematical expressions.
- Reduce computational errors during evaluation.



ABSTARCT

A calculator application is one of the most commonly used software tools for performing arithmetic calculations quickly and accurately. It evaluates mathematical expressions entered by users by following the correct order of operations, including precedence and parentheses. To process these expressions efficiently, an appropriate data structure is required.

Among various data structures, the Stack is the most suitable choice because it follows the Last In, First Out (LIFO) principle. It efficiently stores operators and operands, making it easy to evaluate infix, postfix, and prefix expressions while maintaining the correct execution order. Stack operations such as push and pop require constant time, resulting in fast and efficient expression processing.

Using a stack improves the accuracy, speed, and reliability of calculator applications. It reduces computational errors, simplifies expression evaluation algorithms, and supports complex mathematical calculations. Therefore, the stack data structure is widely used in calculators, programming language compilers, and expression evaluation.